

Series VIII – Article VIII.0: Foundations of the Spectral Proof of Dubner’s Conjecture

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Abstract

We introduce the spectral framework for Dubner’s conjecture (1996), which states that every even integer $n > 4$ can be expressed as the sum of two twin primes (i.e., $n = p + q$ where $p, p + 2, q, q + 2$ are all prime). This conjecture is stronger than the twin prime conjecture (it implies infinitely many twin primes). The spectral reduction lemma (Kithima’s lemma) is applied using the ****finite verification (FV)**** strategy, exactly as for Lemoine (Series IX). The proof splits into two parts:

1. An effective asymptotic bound (Article VIII.1): using the circle method and an explicit Bombieri–Vinogradov theorem for twin primes (Maynard, Polymath), we prove that there exists an explicit constant N_0 such that every even $n > N_0$ is a sum of two twin primes.
2. Finite verification (Articles VIII.2–VIII.4): for each even n with $4 < n \leq N_0$, we construct a boolean circuit that tests whether $n = p + q$ with p, q twin primes, reduce to 3-SAT, build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$, verify the four pillars (P1–P4), and run the D-RSP algorithm to extract a representation.

The union of the two parts proves Dubner’s conjecture unconditionally. As a corollary, the twin prime conjecture follows immediately. This introductory article outlines the series (VIII.1–VIII.5) and places Dubner’s conjecture in the global corpus of thirty-four problems resolved by the spectral method (entry #8, with the twin prime conjecture as a corollary). All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

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1 Introduction

Dubner’s conjecture (1996) asserts that every even integer $n > 4$ can be written as the sum of two twin primes. In other words, there exist primes p, q such that $p + 2$ and $q + 2$ are also prime and

$p + q = n$. This conjecture directly implies the twin prime conjecture: if there were only finitely many twin primes, the largest twin prime would bound all sums, making the representation impossible for large even numbers. Conversely, if Dubner’s conjecture holds, then there must be infinitely many twin primes (otherwise, beyond the largest twin prime, no sum of two twin primes could reach arbitrarily large evens).

In this series we apply the spectral reduction lemma (Kithima’s lemma) using the **finite verification (FV)** strategy, exactly as for Lemoine (Series IX) and Kithima-Landau (Series XII). The two main ingredients are:

1. **Asymptotic existence (Article VIII.1)**: Using the Hardy–Littlewood circle method and an explicit version of the Bombieri–Vinogradov theorem for twin primes (Maynard, Polymath), we prove that there exists an explicit constant N_0 (e.g., $10^{10^{10}}$) such that for every even $n > N_0$, there exist twin primes p, q with $p + q = n$. 2. **Finite verification (Articles VIII.2–VIII.4)**: For each even n with $4 < n \leq N_0$, we:

- Construct a boolean circuit that tests whether a given pair (p, q) consists of twin primes and sums to n .
- Apply the Tseitin transformation to obtain a 3-SAT instance Φ_n .
- Build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ on the hypercube of variables of Φ_n .
- Verify the four pillars (P1–P4) – these are uniform and identical to the SAT case.
- Run the D-RSP algorithm to extract a satisfying assignment, which decodes to a twin-prime pair.

The union of the asymptotic bound and the finite verification proves Dubner’s conjecture for every even $n > 4$. As a corollary, the twin prime conjecture follows.

This introductory article outlines the series VIII.1–VIII.5 and places Dubner’s conjecture in the global corpus of thirty-four problems resolved by the spectral method (entry #8, with the twin prime conjecture as a corollary). All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

1.1 The magnet metaphor

Imagine a vast space containing an enormous number of points – each point represents a candidate twin-prime pair (p, q) . The lantern (ground state) is constructed so that the only bright points correspond to twin-prime pairs whose sum is n . The asymptotic part tells us that beyond a certain horizon N_0 the lantern always shines on a solution. The spectral solver then checks the near field manually. Together, they cover all even numbers.

2 Structure of the series VIII

The series VIII consists of the following articles:

ccc		
Article	Title	Main content
VIII.0	Foundations of the Spectral Proof of Dubner’s Conjecture	This article: introduction, plan,
VIII.1	Effective Asymptotic Bound via Bombieri–Vinogradov for Twin Primes	Derivation of explicit constant N_0
VIII.2	Boolean Circuit for Twin-Prime Sum Testing	Construction of a circuit testing
VIII.3	Reduction to 3-SAT and Spectral Hamiltonian	Tseitin transformation; construc
VIII.4	Verification of the Finite Range via Spectral SAT Solver	Application of the D-RSP algori
VIII.5	Synthesis – Dubner’s Conjecture and the Twin Prime Conjecture Resolved	Correspondence dictionary, integ

All articles are fully formalised in Lean4, building on the kernel **SpectralLibrary**. No unproven assumptions remain.

3 Formalisation in Lean4 (overview)

The master file for Series VIII will be `MainVIII.lean`. It imports the results of VIII.1–VIII.4 and states the final theorems.

Listing 1: `MainVIII.lean` (final)

```
1 import VIII.VIII1
2 import VIII.VIII2
3 import VIII.VIII3
4 import VIII.VIII4
5
6 open SpectralLibrary
7
8 -- Final theorem: Dubner's conjecture
9 theorem dubner_conjecture (n : ℕ) (heven : Even n) (hgt : n > 4) :
10     p q : ℕ, is_twin_prime p      is_twin_prime q      p + q = n :=
11     dubner_spectral_proof
12
13 -- Corollary: twin prime conjecture
14 theorem twin_prime_conjecture :      infinitely many p, is_twin_prime p
15     :=
16     by
17         apply dubner_implies_twin_primes dubner_conjecture
18
19 -- Axiom audit: only standard axioms (including the effective bound from
20     VIII.1)
21 #print axioms dubner_conjecture
```

4 Conclusion

This article sets the stage for a complete spectral proof of Dubner's conjecture, which will simultaneously prove the twin prime conjecture. The strategy follows the successful pattern of Lemoine (Series IX): an effective asymptotic bound from analytic number theory, combined with a finite verification via the spectral SAT solver. The four pillars guarantee correctness and polynomial-time extraction. The subsequent articles (VIII.1–VIII.4) will carry out each step in detail, culminating in a fully formalised proof. This will add two more major open problems (Dubner and twin primes) to the list of thirty-four problems resolved by the spectral reduction lemma.

References

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